5. Link for Infosys Springboard Course:

i) **https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013986385481318400909/overview**

ii) **https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138478721765785605555 2_shared/overview**

iii) **https://infyspringboard.onwingspan.com/web/en/ap/toc/lex_auth_013841728221593600237 0_shared/overview**

Detailed Syllabus:

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignment
1	1. **Rectifiers, Filters and Regulators:** i) Boylestad & Nashelsky (11th edition) Chapter 15	Introduction to full-wave and half-wave rectifiers. Capacitor filter. Inductor filter, LC and π −section filter. Series and Shunt voltage regulator, percentage regulation. Regulator ICs 78xx and 79xx series. Introduction to SMPS.	As per International Standards : (https://explorecourses.stanford.edu/search?q=EE101A) (https://explorecourses.stanford.edu/search?q=EE101B) https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-laboratory-spring-2007/pages/syllabus/ and AICTE-prescribed syllabus: (https://www.aicte-india.org/sites/default/files/Model_Curriculum/Final_ ECE.pdf) ***Industry Mapping:*** LT-SPICE software, LabView, Proteus. Hardware Chipset: IN4001/4007, 7815, 7915.	4	1. Simulation of full-wave and half-wave rectifiers with and without filter using LT-SPICE software to study of Ripple and Regulation characteristics.
	2. **Transistor Biasing and Stability of Q-point:** i) Boylestad & Nashelsky (11th edition)	Introduction to transistor biasing, Q-point selection, Diode Compensation. Bias stabilization and stability factors. h-parameter model of transistors.	As per International Standards : (https://explorecourses.stanford.edu/search?q=EE101A) (https://explorecourses.stanford.edu/search?q=EE101B) https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-	6	2. Simulation of transistor circuit with self bias using LT-SPICE software to study parameters of Q-point. 3. Simulation of transistor amplifier with potential

	Chapter 4 ii) Boylestad & Nashelsky (11th edition) Chapter 5	Analysis of transistor amplifiers, expression for voltage gain, current gain, input and output impedance. Design of transistor amplifier. High frequency model of transistor, short circuit current gain and β cut off frequency.	laboratory-spring-2007/pages/syllabus/ and AICTE-prescribed syllabus : :(https://www.aicte-india.org/sites/default/files /Model_Curriculum/Final_ ECE.pdf) ***Industry Mapping:*** LT-SPICE software, LabView, Proteus. Hardware Chipset: BC547, SL100		divider bias using LT-SPICE software to study output voltage and gain for an input signal.

2	**1. RC coupled amplifier.** i) Boylestad & Nashelsky (11th edition) Chapter 9 **Tuned amplifier:** i) Chattopadhyay and Rakshit Chapter 9	Introduction to RC coupled amplifier, equivalent circuit, analysis of R C coupled amplifier for voltage and current gains, input and output impedances at three frequency ranges, frequency response characteristics, lower and upper cut off frequencies, bandwidth, and concept of wide band amplifier.	As per International Standards : (https://explorecourses.stanford.edu/search?q=EE101A) (https://explorecourses.stanford.edu/search?q=EE101B) https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-laboratory-spring-2007/pages/syllabus/ and AICTE-prescribed syllabus : (https://www.aicte-india.org/sites/default/files /Model_Curriculum/Final_ ECE.pdf) ***Industry Mapping:*** LT-SPICE, LabView, Proteus software. Hardware Chipset: SL 100, BC547.	6	4. Simulation of two stages R-C coupled amplifier using LT-SPICE software to study of its gain and bandwidth.
	2. Feedback Amplifiers & Oscillators: i) Chattopadhyay and Rakshit Chapter 10 and	Introduction to Feedback concept, negative & positive feedback, voltage/current, series/shunt feedback topologies. Introduction to Oscillators, Berkhausen criterion, pole-zero concept, Colpitts , Hartley, Phase	As per International Standards : (https://explorecourses.stanford.edu/search?q=EE101A), (https://explorecourses.stanford.edu/search?q=EE101B), https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-laboratory-spring-2007/pages/syllabus/	4	ii) Simulation of Phase shift oscillator using OPAMP and LT-SPICE software to study frequency of oscillation by varying component values.

	11	shift, Wein bridge and crystal oscillators.	and AICTE-prescribed syllabus : (https://www.aicte-india.org/sites/default/files /Model_Curriculum/Final_ ECE.pdf) ***Industry Mapping:*** LT-SPICE software, LabView, Proteus. Hardware Chipset: SL100, IC 741.		

3	**1. Operational Amplifier:** i) R.A. Gayakwad. Chapter 3	ii) Introduction to operational amplifier, Ideal OP AMP, Differential Amplifier, Constant current source (current mirror etc.), level shifter, CMRR, Open & Closed loop circuits, importance of feedback loop (positive& negative), inverting & non inverting amplifiers, voltage follower/buffer circuit.	As per International Standards : (https://explorecourses.stanford.edu/search?q=EE101A), (https://explorecourses.stanford.edu/search?q=EE101B), https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-laboratory-spring-2007/pages/syllabus/ and AICTE-prescribed syllabus : (https://www.aicte-india.org/sites/default/files /Model_Curriculum/Final_ ECE.pdf), Industry Mapping: LT-SPICE software, LabView, Proteus. Hardware Chipset: IC741, IC328.	6	iii) Simulation of ADC using IC AD570, and using LT-SPICE software. iv) Simulation of DAC using DAC 0808 and using LT-SPICE software. v) Simulation of Inverting and Non Inverting Amplifier using Op-Amp and using LT-SPICE software.
	2. Applications of Operational Amplifiers: i) R.A. Gayakwad. Chapter 6	Applications of Operational Amplifiers: adder, integrator & differentiator, comparator. Instrumentation Amplifier, Log & Anti-log amplifiers, Trans-conductance multiplier, Precision Rectifier, voltage to current and current to voltage converter, free running oscillator.	As per International Standards: (https://explorecourses.stanford.edu/search?q=EE101A) ,(https://explorecourses.stanford.edu/search?q=EE101B), https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-laboratory-spring-2007/pages/syllabus/ and AICTE-prescribed syllabus: (https://www.aicte-india.org/sites/default/files /Model_Curriculum/Final_ ECE.pdf), ***Industry Mapping:*** LT-SPICE	6	vi) Simulation of differentiator and integrator circuit using Op-Amp and using LT-SPICE software. vii) Simulation of Logarithmic and Anti Logarithmic Amplifier circuit using IC TL441, ICL8049 and with

			software, LabView, Proteus. Hardware Chipset: IC741, IC328, Multiplier IC AD633, Logarithmic Multiplier IC TL441, Anti Logarithmic Amplifier ICL 8049.		LT-SPICE software.
4	**Multivibrator:** i) Chattopadhyay and Rakshit Chapter 11	Introduction to multivibrators, Monostable, Bistable, Astable multivibrators using transistor and op-amp; Monostableand astable operation using555 timer.	As per International Standards: (https://explorecourses.stanford.edu/search?q=EE101A), (https://explorecourses.stanford.edu/search?q=EE101B), https://ocw.mit.edu/courses/6-101-introductory-analog-electronics-laboratory-spring-2007/pages/syllabus/ and AICTE-prescribed syllabus : (https://www.aicte-india.org/sites/default/files /Model_Curriculum/Final_ ECE.pdf), ***Industry Mapping:*** LT-SPICE software, LabView, Proteus. Hardware Chipset: IC555.	4	viii) Simulation of Monostable Multivibrator using IC555 and using LT-SPICE software. ix) Simulation of Astable Multivibrator using IC555 and using LT-SPICE software. x) Simulation of Schmitt trigger circuit using IC555 and using LT-SPICE software.

Text book and Reference books:

Text Book:

1. Boylestad, Robert L., and Louis Nashelsky. Electronic devices and circuit theory. PrenticeHall, 2012.
2. Electronics Fundamentals and Applications, D. Chattopadhyay and P. C. Rakshit.
3. Op-amps and Linear IC's, R.A. Gayakwad, PHI.